

# □ BIOLOGY CH.12 — RESPIRATORY SYSTEM — FULL MIND MAP

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## 1. Respiratory System Structure

- Trachea (windpipe) mein C-shaped 20 rings of cartilage hote hain jo air-passage ko collapsing se rokte hain. Ribs thoracic organs protect karti hain, spine body support karti hai. [Q672]
- Respiratory system parts: Nose, Pharynx (throat), Larynx (voice box), Trachea, Bronchi, Bronchioles, Lungs, Alveoli. [Q672, Q681, Q696]
- Pharynx: throat ke peeche, nose/mouth se trachea/oesophagus tak air/food transport. [Q696]
- Larynx (voice box): pharynx ke neeche, vocal cords hain. Sound produce hota hai jab hava cords se guzarti hai. [Q696]
- Trachea: larynx se bronchi tak, air ka passage. [Q696]
- Bronchi: trachea ka divide hona, dono lungs mein jata hai. [Q696]
- ⚠️ Trap: Respiration mein oesophagus included NAHI hota - ye digestive system ka part hai. [Q681]

## 2. Breathing Mechanism & Lungs

- Breathing: physical process of inhaling/exhaling. Cellular respiration: oxygen use karke glucose se energy release karna. [Q695]
- Inspiration (inhalation): intercostal muscles + diaphragm contract → thoracic cavity expand → lungs expand → air andar. [Q675]
- Expiration (exhalation): intercostal muscles + diaphragm relax → thoracic cavity compress → lungs shrink → air bahar. [Q675]
- Lungs thoracic cavity mein hain. Right lung 3 lobes, Left lung 2 lobes (heart ke liye jagah). [Q675]
- Diaphragm: dome-shaped muscle, breathing ka main muscle. Contract = flatten = inhalation. Relax = dome = exhalation. [Q693, Q706]
- Respiratory system ke through body oxygen receive karti hai aur CO<sub>2</sub> remove karti hai. [Q686]

- Pharynx se hava aur dono food ja sakta hai - larynx mein epiglottis food ko trachea mein jaane se rokta hai. [Q704]
- Spirometer se breathing capacity measure hoti hai. [Q690]

### 3. Respiration Process - Cellular

- Respiration ke 3 stages: 1. Glycolysis (cytoplasm): glucose  $\rightarrow$  pyruvate. Oxygen ki zaroorat nahi. 2. Krebs Cycle (mitochondria): pyruvate  $\rightarrow$  acetyl CoA  $\rightarrow$  CO<sub>2</sub> + energy. Oxygen present hona chahiye. 3. Electron Transport Chain (mitochondria): oxygen final electron acceptor, water + ATP banta hai. [Q673]
- ⚠ Trap: Mitochondria mein acetic acid nahi, pyruvic acid breakdown hota hai oxygen use karke. [Q673]
- First stage (glycolysis) cytoplasm mein hota hai - anaerobic. [Q673]
- Oxygen ke presence mein pyruvate mitochondria mein jata hai. [Q673]
- Aerobic respiration: O<sub>2</sub> + Glucose  $\rightarrow$  CO<sub>2</sub> + H<sub>2</sub>O + ATP (mitochondria). [Q708, Q711]

### 4. Gaseous Exchange & Transport

- Alveoli (air sacs) mein gaseous exchange hota hai - lungs ke andar millions of tiny sacs. [Q678]
- Alveoli se O<sub>2</sub> blood mein jata hai, CO<sub>2</sub> blood se alveoli mein aata hai. [Q678]
- Haemoglobin blood mein O<sub>2</sub> bind karke transport karta hai. [Q678]
- Oxygen dissociation curve: haemoglobin ki O<sub>2</sub> binding capacity. Sigmoid shape. [Q678]
- Oxygen blood se tissues mein jaata hai, CO<sub>2</sub> tissues se blood mein aata hai. [Q686]

### 5. Respiration in Plants

- Plants mein stomata, lenticels, root hairs se gas exchange hota hai. [Q683]
- Stomata: mainly leaves mein, guard cells control karte hain. Open = CO<sub>2</sub> in, O<sub>2</sub> out. Close = water conservation. [Q683]
- Lenticels: stems/roots mein, woody plants mein. [Q683]
- Root hairs: soil se oxygen absorb karte hain. [Q683]
- Plants night mein bhi respire karte hain - O<sub>2</sub> lete hain, CO<sub>2</sub> chhodte hain. Day mein photosynthesis zyada hota hai isliye net O<sub>2</sub> release hota hai. [Q683, Q698]
- ⚠ Trap: Plants sirf photosynthesis karte hain, respiration NAHI - ye galat hai. Plants dono karte hain. [Q698]

### 6. Respiration Types - Aerobic & Anaerobic

- Aerobic respiration: Oxygen present, complete glucose breakdown, zyada ATP (~36-38),  $\text{CO}_2 + \text{H}_2\text{O}$  products. [Q687, Q689, Q703, Q708]
- Anaerobic respiration: Oxygen absent, incomplete breakdown, kam ATP (~2), lactic acid/ethanol +  $\text{CO}_2$  products. [Q687, Q689, Q703]
- Human muscles mein exercise ke dauran anaerobic respiration → lactic acid → cramps. [Q689]
- Yeast mein anaerobic respiration → ethanol +  $\text{CO}_2$  (fermentation). [Q689]
- Anaerobic respiration ke products: Lactic acid (animals), Ethanol +  $\text{CO}_2$  (yeast). [Q703]
- Aerobic respiration ke products:  $\text{CO}_2 + \text{H}_2\text{O} + \text{ATP}$ . [Q703, Q711]
- Anaerobic respiration mitochondria mein NAHI hoti - cytoplasm mein hoti hai. [Q708]

## 7. Fermentation

- Fermentation = anaerobic respiration in microorganisms. Yeast: glucose → ethanol +  $\text{CO}_2$ . [Q685, Q699]
- Fermentation mein NADH ko  $\text{NAD}^+$  regenerate kiya jata hai taaki glycolysis chalti rahe. [Q685]
- Fermentation ke types: - Lactic acid fermentation: muscles, some bacteria. - Alcoholic fermentation: yeast, some bacteria. [Q699]
- Fermentation mein oxygen nahi chahiye hoti. [Q699]
- Alcoholic fermentation: yeast, some bacteria. [Q699]

## 8. ATP & Energy

- ATP = Adenosine Triphosphate - cell ki energy currency. [Q674, Q680]
- ATP mein 3 phosphate groups hote hain.  $\text{ADP} + \text{Pi} \rightarrow \text{ATP}$  (energy store).  $\text{ATP} \rightarrow \text{ADP} + \text{Pi}$  (energy release). [Q674, Q680]
- ATP synthesis: substrate-level phosphorylation (glycolysis, Krebs) + oxidative phosphorylation (ETC). [Q674]
- Respiration ka main goal ATP synthesis hai. [Q680]

## 9. Respiratory Disorders

- Asthma: bronchial tubes inflame/contract, mucus zyada, breathing difficult. [Q692]
- Bronchitis: bronchi ka inflammation, mucus zyada. [Q692]
- Emphysema: alveoli walls damage, surface area kam,  $\text{O}_2$  uptake kam. [Q692]
- Pneumonia: lungs infection, alveoli fluid se bharte hain. [Q692]
- Tuberculosis (TB): Mycobacterium tuberculosis se, lungs infect hota hai. [Q692]
- CO poisoning: CO haemoglobin se zyada strongly bind karta hai (carboxyhaemoglobin),  $\text{O}_2$  transport block ho jata hai. [Q710]

## 10. Miscellaneous Facts

- Smoking se lungs damage hote hain - tar deposit, cilia destroy, cancer risk. [Q701]
- Root hairs plants ko water aur minerals absorb karne mein madad karte hain - soil particles se. [Q705]
- Lungs ke andar pleural fluid hota hai jo friction kam karta hai breathing ke dauran. [Q713]
- Tidal volume: normal breath mein air ki amount (~500 mL). Vital capacity: max air exhale kar sakte hain. [Q717]
- Respiratory quotient (RQ):  $\text{CO}_2$  produced/ $\text{O}_2$  consumed. Carbohydrate = 1, Fat = <1, Protein = ~0.8. [Q721]

## 11. Additional Facts — Breathing Mechanics, Gas Exchange & Respiratory Pigment

- Glycolysis (glucose se pyruvate banana) aerobic AUR anaerobic respiration dono ke first step mein common hota hai — cytoplasm mein hoti hai, oxygen ki zaroorat nahi. [Q676]
- 'Breathing holes' (spiracles) Cockroach jaise insects mein milte hain — tracheae network se connected hote hain. Man/Fish alag tarike se saans lete hain. [Q677]
- Pyruvate ka aerobic breakdown human body mein 'Mitochondria' mein hota hai — anaerobically cytoplasm mein hota hai. [Q679]
- Breathing cycle ke dauran lungs mein hamesha bachi rehne wali air 'Residual Volume' kehlaati hai — maximum forceful expiration ke baad bhi kuch air reh jaati hai. [Q682]
- Cigarette smoking harmful hai kyunki ye 'upper respiratory tract ke cilia ko damage karti hai' — airways, alveoli aur cilia (dirt-mucus rokne wale) sabko affect karti hai. [Q684]
- Plants mein respiration ke end products ' $\text{CO}_2$ ,  $\text{H}_2\text{O}$  aur Energy' hote hain —  $\text{Glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Energy}$ , stomata-lenticels-roots se  $\text{O}_2$  lete hain. [Q688]
- Aerobic respiration mein mitochondria mein 1 pyruvate molecule breakdown hokar 3  $\text{CO}_2$  molecules banata hai — aerobic respiration anaerobic se ~18 guna zyada energy release karti hai. [Q691]
- Air human body mein 'Nostrils' ke through enter karti hai — respiration ek metabolic process hai jisme cells  $\text{O}_2$  lekar  $\text{CO}_2$  release karte hain. [Q694]
- Inhaled air mein ~21% Oxygen hoti hai, exhaled air mein ~16% Oxygen — Nitrogen 78% dono mein same rehti hai,  $\text{CO}_2$  inhaled mein 0.04%, exhaled mein 4.4% hoti hai. [Q697]
- Frogs (amphibians) lungs ke alawa apni 'slippery-moist skin' se bhi breathe kar sakte hain (cutaneous respiration). [Q700]
- Kuch fish 'more times per minute' water gulp karti hain animal ki tulna mein jo air breathe karta hai — pani mein dissolved oxygen kam hoti hai, isliye aquatic animals fast breathe karte hain. [Q702]
- 'Sinus' anatomical cavity hai jo air se filled hoti hai — skull ke andar hollow cavities ka connected system. Types: Frontal, Maxillary, Sphenoid, Ethmoid. [Q707]
- 'Cutaneous respiration' (skin se gas exchange) Earthworm mein hoti hai — frogs aur leeches mein bhi hoti hai. [Q709]

- Lungs mein respiratory gases ka exchange 'Alveoli' mein hota hai — tiny air sacs jahan O<sub>2</sub> blood mein jaati hai aur CO<sub>2</sub> diffuse hokar bahar aati hai. [Q712]
- Sea mein green plants photosynthesis se 'Oxygen release' karte hain — ye oxygen paani mein dissolve hokar fish ke gills tak pahunchti hai respiration ke liye. [Q714]
- 'Aerobic Respiration' mein sabse zyada energy release hoti hai — O<sub>2</sub> use karke glucose ka complete breakdown hota hai, 38 ATP tak per glucose molecule. [Q715]
- Muscle cells mein oxygen ki kami hone par 6-carbon glucose molecule '3-carbon' molecule (lactic acid) mein convert ho jaata hai — muscle fatigue-cramps cause karta hai. [Q716]
- Exhalation (expiration) 'lungs ke deflation' se hoti hai — diaphragm-intercostal muscles relax hote hain, thoracic cavity volume kam hota hai, air bahar nikalti hai. [Q718]
- Respiration ke dauran energy 'glucose ke burning' (breakdown) se milti hai — cellular respiration glucose breakdown karke ATP generate karti hai. [Q719]
- Throat (trachea) mein 'Cartilage' ke rings hote hain — collapsing rokhte hain, nose-ears-larynx mein bhi cartilage milti hai. [Q720]
- Human beings mein respiratory pigment 'Hemoglobin' hai — iron-containing pigment, RBC mein milta hai, O<sub>2</sub> ko lungs se body parts tak carry karta hai. [Q722]
- 'Respiration' life process hai jo chemical energy ko heat energy mein convert karta hai — food energy ATP aur heat ki tarah release hoti hai, exothermic reaction hai. [Q723]
- Substrate ka incomplete dissolution oxygen ki absence mein 'Anaerobic Respiration' kehlaata hai — kam energy yield karta hai kyunki electron transport chain less efficient hoti hai. [Q724]
- Sahi statement: 'Oxygen lungs se blood mein diffuse hoti hai, aur Carbon dioxide blood se lungs mein diffuse hoti hai' — alveoli mein O<sub>2</sub> rich hoti hai, blood mein CO<sub>2</sub> zyada. [Q725]
- ⚠ Trap: 'Medulla oblongata' respiratory system ka part NAHI hai — ye BRAIN ka part hai (respiratory rhythm control karta hai, lekin anatomically respiratory system mein nahi aata). Trachea, lungs, bronchi respiratory system ke parts hain. [Q726]
- Anaerobic respiration 'Oxygen' ki absence mein hoti hai — 2 steps: glycolysis aur pyruvic acid ka incomplete breakdown, poisonous compounds jama hone ki wajah se indefinitely continue nahi ho sakti. [Q727]